Worksheet 7.3

Science Quest 9: pages 262–7

Acids and bases

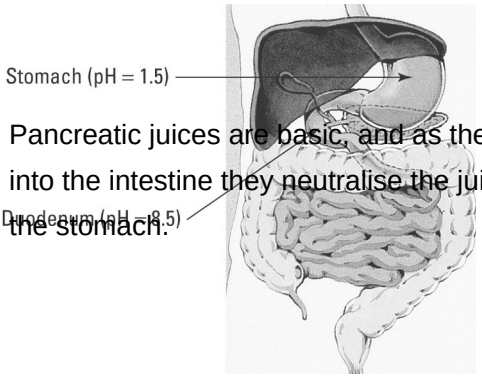
Answers

1. The following table shows the colours of common indicators in solutions of acids and bases. Complete the table for the common household materials.

Indicator	Acid colour	Base colour
Methyl orange	Red–orange	Yellow
Litmus	Red	Blue
Bromothymol blue	Yellow	Blue
Phenolphthalein	Colourless	Pink–crimson

Household material	Lemon juice	Baking powder	Ammonia	Lemonade
Methyl orange	Red	Yellow	Yellow	Red
Litmus	Red	Purplish blue	Blue	Red
Bromothymol blue	Yellow	Blue	Blue	Yellow
Phenolphthalein	Colourless	Pink	Crimson	Colourless

2. The diagram below shows some organs of our digestive system, including the typical pH of the liquids in the stomach and duodenum.



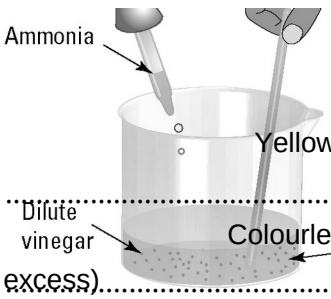
(a) What is the pH of neutral water? 7

(b) Explain why the pH rises as liquids move from the stomach into the duodenum.

Pancreatic juices are basic, and as they pour into the intestine they neutralise the juices from the stomach.

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3. A student adds 5 mL of diluted vinegar to a beaker, and then adds 5 drops of the selected indicator. Drops of household ammonia are then added one at a time while a glass rod is used to stir the mixture.



Describe the observations that would be made if each of the indicators below was used in separate experiments. Assume that an excess of ammonia is finally added to the beaker.

(a) Bromothymol blue:

Yellow to green to blue (in excess)

(b) Phenolphthalein:

Colourless to pink to crimson (in excess)